

# The exact maximal-equilateral number of uncountable $\ell_1(\Gamma)$

Candidate partial-result packet for arXiv:1109.5063

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**Status.** Candidate partial result, likely valid; human review requested.

## 1 Source problem and exact gap

For a normed space  $X$ , Swanepoel and Villa define  $m(X)$  to be the minimum cardinality of a maximal equilateral subset of  $X$  [1]. Their Conjecture 8 asks whether

$$m(X) \leq d + 1$$

for every  $d$ -dimensional normed space  $X$ . On page 4 they prove  $m(\ell_1^d) = 4$  for  $d \geq 3$ , explain why the smooth-point argument fails for uncountable  $\ell_1(\Gamma)$ , and retain only

$$4 \leq m(\ell_1(\Gamma)) \leq 5 \quad (\Gamma \text{ uncountable}).$$

The upper bound is their Theorem 6 (more explicitly, the  $p = 1$  construction in Proposition 19). The result below closes this exact-value gap. It is adjacent progress rather than a solution of the finite-dimensional conjecture.

**Conjecture 8.** *For any  $d$ -dimensional normed space  $X$ ,  $m(X) \leq d + 1$ .*

## 2. A GENERALISATION OF PETTY'S EXAMPLE

Petty [11] showed that  $m(\ell_2^d \oplus_1 \mathbb{R}) = 4$  for all  $d \geq 2$ . In his argument  $\ell_2^d$  can in fact be replaced by any, not necessarily finite-dimensional, normed space which has a *smooth point* on its unit sphere, that is, a point where the norm is Gâteaux differentiable, or equivalently, a point on the unit sphere which has only one supporting hyperplane [6, 12]. By a classical theorem of Mazur [10] any separable normed space enjoys this property [12, Theorem 10].

**Proposition 9.** *Let  $X$  be a normed space of dimension at least 2 with a norm which has a smooth point on its unit sphere. Then  $m(X \oplus_1 \mathbb{R}) = 4$ .*

*Proof.* Since  $X \oplus_1 \mathbb{R}$  is at least 3-dimensional,  $m(X \oplus_1 \mathbb{R}) \geq 4$  by Petty's theorem mentioned in Section 1. For the upper bound, let  $\mathbf{u} \in X$  be a smooth point on the unit sphere of  $X$ . Let  $A := \{(\mathbf{o}, 1), (\mathbf{o}, -1), (\mathbf{u}, 0), (-\mathbf{u}, 0)\}$ . Then  $A$  is a 2-equilateral set in  $X \oplus_1 \mathbb{R}$ . If there exists a point  $(\mathbf{x}, r) \in X \oplus_1 \mathbb{R}$  at distance 2 to each point in  $A$ , then it easily follows that  $r = 0$ ,  $\|\mathbf{x}\| = 1$  and  $\|\mathbf{x} \pm \mathbf{u}\| = 2$ . Then  $\pm \mathbf{x}$ ,  $\pm \mathbf{u}$  and  $\pm \frac{1}{2}\mathbf{x} \pm \frac{1}{2}\mathbf{u}$  are all unit vectors in  $X$  and by convexity the unit ball of the subspace  $Y$  of  $X$  generated by  $\mathbf{u}$  and  $\mathbf{x}$  is the parallelogram with vertices  $\pm \mathbf{u}$  and  $\pm \mathbf{x}$ . In particular, the unit ball of  $Y$  has more than one supporting line at  $\mathbf{u}$ , and so by the Hahn-Banach theorem, the unit ball of  $X$  has more than one supporting hyperplane at  $\mathbf{u}$ .  $\square$

As special cases,  $m(\ell_1) = m(\ell_1^d) = 4$  for  $d \geq 3$ . However, if  $\Gamma$  is an uncountable set, then it is well known that no point on the unit sphere of  $\ell_1(\Gamma)$  is smooth. (This can be seen as follows: Let  $\mathbf{u} \in \ell_1(\Gamma)$  have norm 1. Then  $\mathbf{u}: \Gamma \rightarrow \mathbb{R}$  has countable support  $U \subset \Gamma$ , say. Choose any  $i \in \Gamma \setminus U$ . Then  $\|\mathbf{u} \pm \mathbf{e}_i\|_1 = 2$  and the intersection of the unit ball of  $\ell_1(\Gamma)$  with the subspace generated by  $\mathbf{u}$  and  $\mathbf{e}_i$  is the parallelogram with vertices  $\pm \mathbf{u}$  and  $\pm \mathbf{e}_i$ , as in the proof of Proposition 9.) Nevertheless, Theorem 6 gives the upper bound  $m(\ell_1(\Gamma)) \leq 5$  for any set  $\Gamma$ .

## 3. APPLYING BROUWER'S FIXED POINT THEOREM

## 2 Proof intuition

Four points in an  $\ell_1$  space admit a useful coordinatewise cut decomposition. Sort the four scalar values in each coordinate. The bottom and top gaps act as singleton cuts, while the middle gap acts as one of the three possible 2–2 cuts. Equilateral distances force all four total singleton weights to agree and all three total 2–2 weights to agree.

Now take, in every coordinate, the midpoint of the two middle scalar values. Its distance to a given point is one half of every middle gap, plus precisely the outer gaps assigned to that point. The cut equalities therefore make this a common center. Its radius is strictly smaller than the equilateral side. Four  $\ell_1(\Gamma)$  vectors use only countably many coordinates, so when  $\Gamma$  is uncountable an unused coordinate can supply the missing radius and extend the four-point set.

For the reverse inequality, the five-point Hadamard configuration used in the source at  $p = 1$  is shown directly to remain maximal even when arbitrarily many new coordinates are available.

## 3 The four-point center lemma

**Lemma 1.** *Let  $a_1, a_2, a_3, a_4$  be a  $\lambda$ -equilateral subset of  $\ell_1(\Gamma)$ . There exist  $c \in \ell_1(\Gamma)$  and  $r < \lambda$  such that*

$$\|a_i - c\|_1 = r \quad (i = 1, 2, 3, 4).$$

*Proof.* Fix a coordinate  $\gamma$ . Sort the four real numbers  $a_i(\gamma)$  as

$$b_1 \leq b_2 \leq b_3 \leq b_4$$

and write their consecutive gaps as

$$\alpha_\gamma = b_2 - b_1, \quad \beta_\gamma = b_3 - b_2, \quad \delta_\gamma = b_4 - b_3.$$

If  $\alpha_\gamma > 0$ , exactly one label attains  $b_1$ ; assign this singleton gap to that label. If  $\delta_\gamma > 0$ , assign it to the unique label attaining  $b_4$ . If  $\beta_\gamma > 0$ , the lower two and upper two labels define one of the three unordered partitions

$$12|34, \quad 13|24, \quad 14|23.$$

Zero gaps may be assigned arbitrarily. All gap series below converge, since each coordinate range is at most  $\sum_i |a_i(\gamma)|$ .

Let  $s_i$  be the total outer-gap weight assigned to label  $i$ , and let  $q_{12}, q_{13}, q_{14}$  be the total middle-gap weights of the three displayed partitions. A scalar difference crosses exactly those consecutive gaps whose cuts separate its two labels. Hence

$$\begin{aligned} \|a_1 - a_2\|_1 &= s_1 + s_2 + q_{13} + q_{14}, & \|a_3 - a_4\|_1 &= s_3 + s_4 + q_{13} + q_{14}, \\ \|a_1 - a_3\|_1 &= s_1 + s_3 + q_{12} + q_{14}, & \|a_2 - a_4\|_1 &= s_2 + s_4 + q_{12} + q_{14}, \\ \|a_1 - a_4\|_1 &= s_1 + s_4 + q_{12} + q_{13}, & \|a_2 - a_3\|_1 &= s_2 + s_3 + q_{12} + q_{13}. \end{aligned}$$

Equality of complementary pair distances first gives

$$s_1 + s_2 = s_3 + s_4, \quad s_1 + s_3 = s_2 + s_4, \quad s_1 + s_4 = s_2 + s_3.$$

Thus  $s_1 = s_2 = s_3 = s_4 =: s$ . Comparing the remaining pair distances then gives

$$q_{12} = q_{13} = q_{14} =: q.$$

Consequently the common side length is

$$\lambda = 2s + 2q. \tag{1}$$

Define

$$c(\gamma) := \frac{b_2 + b_3}{2}.$$

This defines an element of  $\ell_1(\Gamma)$  because  $|c(\gamma)| \leq \max_i |a_i(\gamma)| \leq \sum_i |a_i(\gamma)|$ . At coordinate  $\gamma$ , every label pays  $\beta_\gamma/2$  to reach  $c(\gamma)$ ; the unique bottom label additionally pays  $\alpha_\gamma$ , and the unique top label additionally pays  $\delta_\gamma$ . Therefore

$$\|a_i - c\|_1 = s_i + \frac{1}{2}(q_{12} + q_{13} + q_{14}) = s + \frac{3}{2}q =: r$$

for every  $i$ . By (1),

$$\lambda - r = s + \frac{1}{2}q > 0,$$

where strictness follows from  $\lambda > 0$  and  $s, q \geq 0$ . □

## 4 Exact value for uncountable $\ell_1(\Gamma)$

**Theorem 2.** *For every uncountable set  $\Gamma$ ,*

$$\boxed{m(\ell_1(\Gamma)) = 5.}$$

*Proof.* First consider a four-point  $\lambda$ -equilateral set  $A = \{a_1, a_2, a_3, a_4\}$ . By Lemma 1, choose  $c$  and  $r < \lambda$  equidistant from its points. The union of the supports of  $a_1, \dots, a_4$  is countable, and the construction of  $c$  uses no other coordinates. Choose  $\gamma_0 \in \Gamma$  outside this union and put

$$x = c + (\lambda - r)e_{\gamma_0}.$$

The new coordinate is disjoint from every  $a_i - c$ , so

$$\|x - a_i\|_1 = \|c - a_i\|_1 + \lambda - r = \lambda$$

for all  $i$ . Thus no four-point equilateral set is maximal. The source recalls Petty's theorem that in dimension at least three every equilateral triple extends to four points [1]; pairs extend to triples by the standard continuity argument. Hence  $m(\ell_1(\Gamma)) \geq 5$ .

For the matching upper bound, choose four distinct coordinates of  $\Gamma$  and, suppressing all remaining zero coordinates, set

$$\begin{aligned} p_1 &= (1, 1, 1, 0), & p_2 &= (1, -1, -1, 0), \\ p_3 &= (-1, 1, -1, 0), & p_4 &= (-1, -1, 1, 0), \\ p_5 &= (0, 0, 0, 1). \end{aligned}$$

All ten pairwise  $\ell_1$  distances are 4.

Suppose that  $x = (x_1, x_2, x_3, x_4, z)$  were at distance 4 from all five points, where  $\rho = \|z\|_1$  accounts for all remaining coordinates. Put

$$h(t) = |t - 1| - |t + 1|.$$

Equality of the distances from  $x$  to  $p_1, p_2, p_3, p_4$  gives

$$h(x_2) + h(x_3) = 0, \quad h(x_1) + h(x_3) = 0, \quad h(x_1) + h(x_2) = 0.$$

These equations force  $h(x_1) = h(x_2) = h(x_3) = 0$ , and  $h(t) = 0$  only for  $t = 0$ . Hence  $x_1 = x_2 = x_3 = 0$ . The distance to any of the first four points now yields

$$3 + |x_4| + \rho = 4, \quad \text{so } |x_4| + \rho = 1.$$

But then

$$\|x - p_5\|_1 = |x_4 - 1| + \rho \leq |x_4| + 1 + \rho = 2 < 4,$$

a contradiction. Thus the five-point set is maximal and  $m(\ell_1(\Gamma)) \leq 5$ . □

## 5 Verification, scope, and novelty

The script `code/verify.py` solves the cut-weight linear system symbolically. It confirms

$$s_1 = s_2 = s_3 = s_4, \quad q_{12} = q_{13} = q_{14}, \quad \lambda - r = s + \frac{q}{2}.$$

It also checks exact rational permutation-symmetrized examples with three nonzero gaps per coordinate, the unused-coordinate extension, and all ten distances in the maximal five-point set.

This theorem determines the exact value left between the source’s lower and upper observations for uncountable  $\ell_1(\Gamma)$ . It does not resolve the finite-dimensional Conjecture 8.

Cheap run indexes were searched for arXiv:1109.5063 and the core maximal-equilateral terminology. Bounded web searches on 21 July 2026 used exact variants of “ $m(\ell_1(\Gamma)) = 5$ ”, “uncountable  $\ell_1$  maximal equilateral”, and four-point  $L_1$  cosphericity. They found the source, related infinite-dimensional work, and an expository five-point noncospherical example, but no later proof of this exact value or the four-point lemma. This is not an exhaustive novelty certification.

**Human-review recommendation.** Check the cut bookkeeping at tied scalar values, convergence of the nonnegative gap sums, and the strict radius calculation. The five-point maximality argument is independent and elementary.

## References

- [1] K. J. Swanepoel and R. Villa, *Maximal equilateral sets*, Discrete Comput. Geom. **50** (2013), 354–373; arXiv:1109.5063.